

CMP513 - Computer Graphics and Image Processing

Tutorial Activity

Descriptive questions

1. **Types of Matrices:** Describe and provide an example for each of the following types of matrices:

- Symmetric Matrix
- Skew-Symmetric Matrix
- Diagonal Matrix
- Scalar Matrix
- Identity Matrix

2. **Matrix Addition and Subtraction:** Explain the conditions under which two matrices can be added or subtracted. Given two matrices, calculate $A+B$ and $A-B$.

$$A = \begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix} \text{ and } B = \begin{bmatrix} -1 & 0 \\ 5 & 2 \end{bmatrix}$$

3. **Matrix Multiplication:** What are the conditions for the product of two matrices A and B to be defined (i.e., for AB to exist)? Is matrix multiplication commutative? Justify your answer with an example.

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \text{ and } B = \begin{bmatrix} 0 & 1 \\ 2 & 3 \end{bmatrix}$$

4. **Determinant of a Matrix:** Explain the concept of the determinant of a square matrix. How is the determinant used to determine if a matrix has an inverse? Calculate the determinant of the matrix

$$C = \begin{bmatrix} 3 & 2 \\ 1 & 4 \end{bmatrix}$$

5. **Inverse of a Matrix:** Define the inverse of a square matrix. What is a singular matrix, and why does it not have an inverse? Find the inverse of the matrix

$$D = \begin{bmatrix} 2 & 1 \\ 5 & 3 \end{bmatrix}$$

Multiple choice questions

1. A matrix where all elements above the main diagonal are zero is called a:
 - a) Diagonal matrix
 - b) Identity matrix
 - c) Lower triangular matrix
 - d) Upper triangular matrix
2. If matrix A has dimensions $m \times n$ and matrix B has dimensions $p \times q$, for the product AB to be defined, which condition must be met?
 - a) $m=p$
 - b) $n=q$
 - c) $n=p$
 - d) $m=q$
3. Which of the following is true for a symmetric matrix A?
 - a) $A^T = -A$
 - b) $A^T = A$
 - c) $A^2 = I$
 - d) $\det(A) = 0$
4. Given matrices $P = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$ and $Q = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, what is $P+Q$?
 - a) $\begin{bmatrix} 1 & 4 \\ 3 & 4 \end{bmatrix}$
 - b) $\begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$
 - c) $\begin{bmatrix} 0 & 3 \\ 2 & 0 \end{bmatrix}$
 - d) $\begin{bmatrix} 1 & 4 \\ 1 & 4 \end{bmatrix}$
5. If $A = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$, then $3A$ is:
 - a) $\begin{bmatrix} 6 & 0 \\ 0 & 2 \end{bmatrix}$
 - b) $\begin{bmatrix} 2 & 0 \\ 0 & 6 \end{bmatrix}$
 - c) $\begin{bmatrix} 6 & 0 \\ 0 & 6 \end{bmatrix}$
 - d) $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$
6. For a square matrix A, if $AA^{-1} = I$, where I is the identity matrix, then A^{-1} is the:
 - a) Transpose of A
 - b) Adjoint of A
 - c) Inverse of A
 - d) Determinant of A
7. A square matrix A is invertible if and only if:
 - a) $\det(A) = 0$
 - b) $\det(A)^{-1} = 0$
 - c) A is a null matrix
 - d) A is a diagonal matrix
8. The determinant of the matrix $\begin{bmatrix} 5 & 2 \\ 3 & 1 \end{bmatrix}$ is:

- a) 11
 - b) -1
 - c) 7
 - d) 1
9. If A is a 3×3 matrix and $\det(A)=5$, then $\det(A^{-1})$ is:
- a) 5
 - b) -5
 - c) $1/5$
 - d) 0
10. Which of the following statements is true regarding matrix multiplication?
- a) $AB=BA$ always
 - b) $(AB)^T=A^TB^T$
 - c) $(AB)^{-1}=B^{-1}A^{-1}$
 - d) $A(B+C)=AB+CA$